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● *Clinical Original Contribution*

EPISCLERAL PLAQUE RADIOTHERAPY IN THE TREATMENT OF UVEAL MELANOMAS

ZBIGNIEW PETROVICH, M.D.,* GARY LUXTON, PH.D.,* BRYAN LANGHOLZ, PH.D.,† MELVIN A. ASTRAHAN, PH.D.* AND PETER E. LIGGETT, M.D.‡

University of Southern California School of Medicine, 1441 Eastlake Ave., Los Angeles, CA 90033

During an 8-year period, 85 patients with uveal melanomas were treated with episcleral plaque radiotherapy (EPRT). The T-stage was: T_1 – 3 (4%), T_2 – 29 (34%) and T_3 – 53 (62%). The mean tumor elevation was 6.1 mm. Radiation dose was prescribed at the tumor apex and at $D_{5\,mm}$. The mean $D_{5\,mm}$ dose was 150.1 Gy (range 70.5–430 Gy) and the mean dose at the apex was 102.6 Gy (range 29.8–200 Gy). Useful vision (> 5/200) was maintained in 73% of patients. The 5-year actuarial survival was 88%. Metastatic disease developed in 9 (11%) patients, 6 of whom died of their disease. Basal tumor dimensions were important factors predicting metastatic disease, $p = 0.002$. A decrease in tumor elevation was seen in 82%. There was a much lower incidence of decrease in tumor radial and circumferential dimensions, 47.5 and 46%, respectively, $p < 0.001$. Treatment complications were common (56%), particularly in patients with large tumors (72%), $p = 0.04$. The incidence of complications was higher in patients treated prior to 1988 as compared to those who were treated more recently (67 vs 35%, $p = 0.010$). There were 13 (15%) patients who had enucleation. This included 12 treated before 1986 and 1 patient treated subsequently (46 vs 2%, $p < 0.001$). In a univariate analysis, tumor height and radiation dose at $D_{5\,mm}$ were important factors predicting enucleation, $p = 0.004$. In a multivariate analysis, however, the most important factor predicting enucleation was treatment administration prior to 1986, $p < 0.001$). A sharp decrease in the incidence of severe complications, including enucleation, as seen after 1985, is likely due to a major effort in treatment optimization.

Uveal melanoma, Episcleral radiotherapy.

INTRODUCTION

Treatment of patients with uveal melanoma has undergone an evolution. Until the early 1970's, surgical management, consisting primarily of enucleation, was the only widely recognized and effective treatment for these patients (3, 6, 23). Radiation therapy with episcleral radioactive plaques (EPRT) or particle beams in the past 20 years have become a treatment of equal importance to surgery for T_1, T_2 and selected T_3 lesions (1, 3, 6, 7, 13). The attractiveness of radiotherapy is due to its conservative nature. This treatment results in excellent survival with a preservation of useful vision in 50 to 70% of patients (4, 13, 16, 19). External photon beam radiotherapy has also been widely applied as an adjuvant treatment used prior to enucleation in larger T_3 tumors. The rationale behind this use of radiotherapy was an attempt to reduce the incidence of postsurgical metastatic disease (8, 9, 12, 23).

In 1985, the Collaborative Ocular Melanoma Study (COMS) was formed. The main objective of this group was to define optimal treatment for patients with uveal melanomas. This was to be accomplished through the use of multi-institutional prospective randomized trials. These trials are well underway and, in several years, they are expected to provide important data to assist ophthalmic oncologists in the selection of an appropriate therapy for patients with choroidal melanoma.

The purpose of this report is to present our clinical experience with EPRT for selected patients with choroidal melanoma.

METHODS AND MATERIALS

From 1983 to 1990, 85 patients with choroidal melanoma received EPRT at the University of Southern California School of Medicine (USC). An additional 21 patients were treated in a Phase I trial with episcleral thermoradiotherapy, and 12 patients with medium sized tumors were randomized to COMS EPRT versus enucleation, while 11 were randomized to COMS preoperative radiotherapy versus enucleation alone. Since 1985, only patients who were not eligible for COMS studies or who

* Dept. of Radiation Oncology.
† Dept. of Preventive Medicine.
‡ Dept. of Ophthalmology.

Reprint requests to: Zbigniew Petrovich, M.D.
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